

POKHARA UNIVERSITY

Level : Bachelor  
Program: BE  
Course : Artificial Intelligence

Semester – Spring

Year : 2020

Full Marks : 70

Pass Marks : 31.5

Time : 2 hrs

Candidates are required to answer in their own words as far as practicable.  
The figures in the margin indicate full marks.

Attempt all the questions.

Section– A: (5×10 = 50)

1. "Rational Thinking" as an aspect of Artificial Intelligence (AI), explain. Discuss on the various application domains of AI. 5+5
2. Mention search criteria. How are informed search methods more efficient than uninformed search? Explain the algorithm of A\* search with a suitable example. 2+2+6
3. Discuss Propositional logic with its complexity, validity and tautology. Convert the following sentences into FOPL sentences. 5+5
  - a) No mango is blue.
  - b) You can fool some of the people all the times
  - c) Everyone is younger than his father.
  - d) None of the gardener likes sun.
  - e) All purple mushrooms are venomous.
4. Differentiate Semantic net and Frame. Construct a parse tree for the sentence "The students took the online examination". 4+6
5. Differentiate Supervised Learning with Unsupervised learning. Discuss K-means clustering with following dataset with k=2. 4+6

| StaffID | Age | NO of leaves |
|---------|-----|--------------|
| 101     | 20  | 4            |
| 102     | 29  | 7            |
| 103     | 35  | 6            |
| 104     | 22  | 9            |
| 105     | 28  | 8            |

OR

Define Neural Network (NN). How back propagation algorithm is used in learning and predicting data in machine learning? Explain with suitable example. 2+8

Section – B: (1×20 = 20)

6. a) What do you mean by production system? Solve the following Crypto arithmetic problem. 2+8

COCA  
+ COLA  
PEPSI

- b) Introduce statistical reasoning and state Bayes Theorem.

A Scam detector system detects spam emails. Assume that the word "offer" occurs in 80% of the spam messages in your account and in 10% of your non-spam emails (i.e. genuine). 30% of the received emails are considered as a scam. Now, you will receive a new message which contains the word "offer". What is the probability that it is a spam?